

NILAY SINGH

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EDUCATION

Vellore Institute of Technology

September 2022 – August 2026

Btech in Computer Science and Engineering

PROFESSIONAL SUMMARY

Aspiring Data Analyst with strong skills in Python, SQL and Excel. Experienced in data cleaning, exploratory data analysis (EDA), and building insightful visualizations to solve business problems. Passionate about transforming raw data into actionable insights.

TECHNICAL SKILLS

Languages: Python, SQL, Excel, CSV, HTML, CSS , Java

Libraries/Frameworks: Pandas, NumPy, mysql.connector, pymysql, SQLAlchemy

Data Visualization: Matplotlib, Seaborn

Developer Tools: MS Excel, Power BI, VS Code, Python IDLE, Jupiter Notebook, Google Collaboratory, MySQL Workbench, MySQL Command Line Client

Databases: MySQL

Concepts: Database Management, Data Cleaning, Exploratory Data Analysis (EDA), Data Visualization, Data Interpretation, Cloud Computing

PROJECTS

AirBnB Booking Analysis | Python, Pandas, Matplotlib, Seaborn, EDA, Data Visualization

- Conducted exploratory data analysis on **Airbnb listings dataset(102599 listings)** to uncover pricing patterns and customer booking trends using **Python**.
- Performed data cleaning, handling missing values, and feature analysis using **Pandas and NumPy**.
- Built data visualizations (heatmaps, distribution plots, correlation analysis) to identify relationships between price, location, room type, and availability.
- Generated actionable insights that could help optimize listing pricing and improve booking performance. Answered **5+ Analytial Questions** and created **8+ Visualizations**

Smart Transaction Monitoring System | Python, MySQL, Pandas, Matplotlib

- Developed a rule-based system to detect suspicious financial transactions using conditions such as high transaction amount and transaction frequency.
- Implemented logic to flag transactions exceeding defined thresholds (e.g., amount greater than 10,000 and multiple transactions within a short time period).
- Integrated MySQL database to store and manage transaction records, enabling efficient querying and analysis.
- Utilized Pandas for data processing, cleaning, and transformation of transaction data.
- Visualized transaction patterns and suspicious activity using Matplotlib to gain actionable insights.
- Designed a simple interface to simulate real-time transaction monitoring.

Trainfo | Python, MySQL

- Designed and developed a desktop-based ticket reservation system that enables users to book tickets, view reservations, check train routes, and search available trains through an intuitive GUI.
- Implemented a **MySQL** relational database to manage trains, routes, users, and booking records, ensuring data integrity and efficient query performance.
- Built the application backend using **Python and SQL**, integrating **mysql.connector**, **pymysql**, and **SQLAlchemy** for secure and efficient database connectivity.
- Utilized **Pandas** for data processing, validation, and transformation of train schedules and reservation data.
- Created a user-friendly interface using **EasyGUI**, allowing non-technical users to interact with the system easily.

CERTIFICATIONS AND ACHIEVEMENTS

- HackerRank- SQL (Basic) Certificate
- Python Essentials - Vityarthi
- The Bits and Bytes of Computer Networking - Coursera
- Solved 20+ SQL/database problems on **LeetCode**, improving data querying and problem-solving skills.
- 4 Star SQL - HackerRank
- Cloud Computing - NPTEL